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### AEROSOL GENERATING ARTICLE AND AEROSOL GENERATING SYSTEM

#### Abstract

An aerosol generating article includes an aerosol generating rod, which includes an aerosol generating material, and a filter rod arranged downstream of the aerosol generating rod and including a capsule. The capsule includes a core region, a first shell, and a second shell, and the core region includes a phenol reduction material.

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## Background/Summary

### CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority to and the benefit of Korean Patent Application No. 10-2024-0031531, filed on Mar. 5, 2024, and Korean Patent Application No. 10-2025-0008184, filed on Jan. 20, 2025, the contents of which are incorporated by reference in their entireties.

### BACKGROUND

#### 1. Field

[0002] The disclosure relates to an aerosol generating article and an aerosol generating system, and more particularly, to an aerosol generating article and an aerosol generating system including a phenol reduction material.

#### 2. Description of the Related Art

[0003] Recently, there is an increasing demand for alternative methods of overcoming shortcomings of general cigarettes. For example, there is an increasing demand for a method of generating aerosol by heating an aerosol generating material in cigarettes, rather than by burning cigarettes. Accordingly, research and development on heated cigarettes or heated aerosol generating devices have been in active progress.

[0004] When manufacturing filters for aerosol generating articles, phenolic smoke ingredients in mainstream smoke have been reduced by adding phenol reduction materials, by which phenols generated during smoking may be specifically reduced. In the related art, it is known that polyethylene glycol (PEG), triethyl citrate (TEC), triacetin (TA), and the like, as phenol reduction materials, have been added by being applied onto filters for aerosol generating articles in the form of a spray type, e.g., through a transfer jet nozzle system (TJNS).

### SUMMARY

[0005] When a phenol reduction material is added by being applied onto a filter for an aerosol generating article, as the phenol reduction material is depleted in proportion to a storage period of the aerosol generating article, there may be issues regarding storage stability.

[0006] The disclosure provides an aerosol generating article and an aerosol generating system, by which phenol smoke ingredients may be reduced as much as possible by securing the storage stability of the phenol reduction material by using a capsule including double shells.

[0007] Technical goals to be achieved based on embodiments are not limited to the technical goals mentioned above, and the technical goals not mentioned above may be clearly understood to those skilled in the art through the present specification and the following drawings.

[0008] Additional aspects will be set forth in part in the description which follows and, in part, will be apparent from the description, or may be learned by practice of the presented embodiments of the disclosure.

[0009] According to an embodiment, an aerosol generating article may include an aerosol generating rod, which includes an aerosol generating material, and a filter rod arranged downstream of the aerosol generating rod and including a capsule. The capsule may include a core region, a first shell, and a second shell, and the core region may include a phenol reduction material.

[0010] According to another embodiment, an aerosol generating system may include an aerosol generating article including an aerosol generating rod and a filter rod, the aerosol generating rod including an aerosol generating material and the filter rod arranged adjacent to the aerosol generating rod and including a capsule, an insertion space for insertion of the aerosol generating

article, and a heater arranged in correspondence to the aerosol generating rod around the insertion space. The capsule may include a core region, a first shell, and a second shell, and the core region may include a phenol reduction material.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

[0011] The above and other aspects, features, and advantages of certain embodiments of the disclosure will be more apparent from the following description taken in conjunction with the accompanying drawings, in which:

[0012] FIG. 1 is a block diagram of an aerosol generating device according to an embodiment;

[0013] FIG. 2 illustrates an aerosol generating device according to an embodiment;

[0014] FIG. 3 illustrates an aerosol generating device according to an embodiment;

[0015] FIG. 4 illustrates an aerosol generating device according to an embodiment;

[0016] FIGS. 5 and 6 are diagrams illustrating examples of an aerosol generating article;

[0017] FIG. 7 is a diagram illustrating a capsule included in an aerosol generating article according to an embodiment; and

[0018] FIG. 8 is a diagram for illustrating an aerosol generating system according to an embodiment.

### DETAILED DESCRIPTION

[0019] Hereinafter, embodiments disclosed herein will be described in detail with reference to the accompanying drawings, and regardless of drawing symbols, same reference numerals will be given to same or similar elements, and descriptions thereof will not be repeatedly given. Regarding descriptions with reference to the drawings, similar drawing symbols may be used for similar or related elements.

[0020] Words such as “module” or “unit” for elements, which will be used in the following descriptions, are given or interchangeably used only in consideration of the ease of drafting the specification, and the words themselves do not have meanings or functions distinguished from each other. The words “module” or “unit” may include units implemented as hardware, software, or firmware, and for example, may be compatibly used with the term such as logics, logic blocks, components, or circuits. “Module” or “unit” may indicate an integrally configured component, or may indicate a minimal unit or a portion of the component performing one or more functions. For example, “module” or “unit” may be implemented in the form of an Application-Specific Integrated Circuit (ASIC).

[0021] In the descriptions of the embodiments disclosed herein, when it is determined that detailed descriptions with respect to related public technologies may make unclear of the main idea of the embodiments disclosed herein, detailed descriptions of the embodiments will be omitted. In addition, it will be understood that the accompanying drawings are only used for the ease of understanding about the embodiments disclosed herein and are not to limit the technical idea disclosed herein, and the accompanying drawings enclose all modifications, equivalents, and substitutions included in the technical spirit and the scope of the disclosure.

[0022] Although terms including ordinal numbers such as “first” and “second” may be used for describing various type of elements, but the elements are not limited to the terms. The terms are only used to distinguish one element from others.

[0023] When an element is mentioned as being “connected to” or “in contact with” another element, it will be understood that the element is directly connected to or in contact with the other element, but another element may be therebetween. When an element is mentioned as being “directly connected to” or “directly in contact with” another element, it will be understood that there is no another element therebetween.

[0024] Unless explicitly indicated otherwise in context, singular expressions also encompass plural expressions.

[0025] Embodiments may be implemented as software including one or instructions stored in a machine (e.g., an aerosol generating device **1**)-readable storage medium (e.g., a memory **17**). For example, a processor (e.g., a controller **12**) of the machine (e.g., the aerosol generating device **1**) may be configured to call at least one command from among the one or more instructions stored in the storage medium and execute the at least one command. By doing so, the machine may be operated to perform at least one function in response to the at least one instruction that has been called. The one or more instructions may include code generated by a compiler or code executable by an interpreter. The machine-readable storage medium may be provided in the form of a non-transitory storage medium. Here, ‘non-transitory’ only indicates that the storage medium is a tangible device and does not include signals (e.g., electromagnetic waves), and the term ‘non-transitory’ does not distinguish a case where data is semi-permanently stored in the storage medium and a case where data is temporarily stored in the storage medium.

[0026] In the disclosure, directions of the aerosol generating device **1** may be defined based on an orthogonal coordinate system. In the orthogonal coordinate system, a x-axis direction may be defined as a horizontal direction of the aerosol generating device **1**. A y-axis direction may be defined as a longitudinal direction of the aerosol generating device **1**. A z-axis direction may be defined as a vertical direction of the aerosol generating device **1**.

[0027] FIG. **1** is a block diagram of the aerosol generating device **1** according to an embodiment.

[0028] According to an embodiment, the aerosol generating device **1** may include a power source **11**, the controller **12**, a sensor unit **13**, an output unit **14**, an input unit **15**, a communication unit **16**, the memory **17**, and heaters (e.g., a heater **18** and a cartridge heater **24**). However, it will be understood to those skilled in the art that, according to the design of the aerosol generating device **1**, some of the elements shown in FIG. **1** may be omitted, or other elements may be added.

[0029] According to an embodiment, the sensor unit **13** may be configured to sense the state of the aerosol generating device **1** or the state around the aerosol generating device **1** and deliver sensed information to the controller **12**. For example, the sensor unit **13** may include a temperature sensor, a puff sensor, an insertion sensor, a reuse sensor, a humidity sensor, a cigarette recognition sensor, a cartridge sensor, a cap sensor, and/or a motion sensor. The sensor unit **13** may further include various types of sensors, e.g., a fluid level sensor configured to sense a fluid level in the cartridge, a submersion sensor configured to sense submersion of the aerosol generating device **1**.

[0030] According to an embodiment, the temperature sensor may be configured to sense temperatures to which the heaters (i.e., the heater **18** and the cartridge heater **24**) are heated. The aerosol generating device **1** may further include another temperature sensor configured to sense temperature of the heaters (i.e., the heater **18** and the cartridge heater **24**), or the heaters (i.e., the heater **18** and the cartridge heater **24**) themselves may function as temperature sensors. For example, the temperature sensor may be used to measure an impedance of the heater **18**. The impedance of the heater **18** may have a correlation relationship with the temperature of the heater **18**. The temperature sensor may be configured to measure a current and/or a voltage applied to the heater **18** (or an induction coil). Based on the current and/or the voltage measured, the impedance of the heater **18** may be calculated. The controller **12** may be configured to estimate the temperature of the heater **18** based on the impedance that has been calculated.

[0031] For example, the temperature sensor may include a resistor (e.g., a thermistor) whose resistance value changes in correspondence to changes in the temperature of the heaters (i.e., the heater **18** and the cartridge heater **24**). The temperature sensor may be configured to output a signal corresponding to the resistance value of the resistor, and the controller **12** may be configured to detect the temperature and/or change in the temperature, based on the signal corresponding to the resistance value.

[0032] As another example, the temperature sensor may include a sensor configured to detect

resistance values of the heaters (i.e., the heater **18** and the cartridge heater **24**). The temperature sensor may be configured to output signals corresponding to the resistance values of the heaters (i.e., the heater **18** and the cartridge heater **24**), and the controller **12** may be configured to detect the temperatures and/or changes in the temperatures, based on the signals corresponding to the resistance values.

[0033] According to an embodiment, the temperature sensor may be configured to sense a temperature of the power source **11**. The temperature sensor may be arranged adjacent to the power source **11**. For example, the temperature sensor may be attached to a surface of the power source **11** (e.g., a battery) and/or mounted onto a surface of a printed circuit board. For example, the aerosol generating device **1** may include a protection circuit module, and the temperature sensor may be arranged adjacent to the power source **11**, together with the protection circuit module PCM.

[0034] According to an embodiment, the temperature sensor may be arranged in a housing (not shown) of the aerosol generating device **1** and sense a temperature in the housing (not shown).

[0035] According to an embodiment, a puff sensor may be configured to sense puffs of a user.

[0036] For example, the puff sensor may include a pressure sensor. The pressure sensor may be configured to output a signal corresponding to an internal pressure of the aerosol generating device **1**, and the controller **12** may be configured to detect puffs of the user, based on the signal corresponding to the internal pressure. Here, the internal pressure of the aerosol generating device **1** may correspond to a pressure of an airflow path through which air flows. The puff sensor may be arranged in the aerosol generating device **1**, in correspondence to the airflow path through which the air flows.

[0037] As another example, the puff sensor may include the temperature sensor. When the puffs of the user occur, temperatures may temporarily drop in the airflow path, a space into which the aerosol generating article is inserted (e.g., an insertion space), the heaters (i.e., the heater **18** and the cartridge heater **24**), and the like. The controller **12** may be configured to detect the puffs of the user, based on signals corresponding to the temperature of the airflow path and the like output from the temperature sensor.

[0038] As another example, the puff sensor may include both of the pressure sensor and the temperature sensor. In this case, the temperature sensor may be configured to measure a temperature used for calibrate the internal pressure measured by the pressure sensor. For example, the puff sensor may be configured to calibrate the signal corresponding to the internal pressure, based on the temperature measured by the temperature sensor, and output the signal that has been calibrated. As another example, the puff sensor may be configured to output a signal corresponding to the temperature measured by the temperature sensor and the signal corresponding to the internal pressure measured by the puff sensor. In this case, the controller **12** may receive the aforementioned signals, and may calibrate the signal corresponding to the internal pressure based on the signal corresponding to the temperature.

[0039] As another example, the puff sensor may include a capacitance sensor. In the disclosure, the capacitance sensor may also be referred to as a cap sensor or a capacitive sensor. When the puffs of the user occur, change in the temperature and/or flow of aerosol may occur in the insertion space of the aerosol generating article, and accordingly, permittivity in the insertion space may change. The controller **12** may be configured to detect the puffs of the user, based on signals corresponding to the permittivity in the insertion space and the like output from the capacitance sensor.

[0040] The puff sensor is not limited to the aforementioned examples, and may be implemented as various types of sensors for sensing the puffs of the user.

[0041] According to an embodiment, the insertion sensor may be configured to sense insertion and/or removal of the aerosol generating article. The insertion sensor may be arranged adjacent to the insertion space. In addition, the insertion sensor may include arbitrary combinations of the aforementioned examples.

[0042] For example, the insertion sensor may include the capacitance sensor. The capacitance

sensor may include at least one conductor, and the at least one conductor may be arranged adjacent to the insertion space. When the aerosol generating article is inserted into/removed from the insertion space, permittivity near the conductor may change. The controller **12** may be configured to detect insertion and/or removal of the aerosol generating article, based on the signals corresponding to permittivity and the like output from the capacitance sensor.

[0043] As another example, the insertion sensor may include an inductive sensor. The inductive sensor may include at least one coil, and the at least one coil may be arranged adjacent to the insertion space. In a case where the aerosol generating article (e.g., a wrapper in the aerosol generating article) includes the conductor, when the aerosol generating article is inserted into/removed from the insertion space, changes in a magnetic field may occur around a coil through which a current flows. The controller **12** may be configured to sense insertion and/or removal of the aerosol generating article including the conductor, based on properties (e.g., frequencies, current values, voltage values, inductance, impedance and the like of an alternating current) of the current output from the inductive sensor or sensed by the inductive sensor. Alternatively, a susceptor and the like may be included in the aerosol generating article (e.g., an aerosol generating rod of the aerosol generating article). In this case, changes in the magnetic field may occur around the coil, based on insertion or removal of the susceptor into/from the insertion space, and the controller **12** may sense insertion and/or removal of the aerosol generating article based on the properties of the current of the inductive sensor.

[0044] The insertion sensor is not limited to the aforementioned examples, and may also be implemented as various types of sensors (e.g., a proximity sensor) configured to sensing insertion and/or removal of the aerosol generating article. In addition, the insertion sensor may include arbitrary combinations of the aforementioned examples. According to an embodiment, the insertion sensor may also include a switch and the like configured to sensing pressure by the aerosol generating article.

[0045] According to an embodiment, the reuse sensor may be configured to sense whether the aerosol generating article is reused. For example, the reuse sensor may include a color sensor configured to sense color of the aerosol generating article. When the aerosol generating article is used by the user, due to generated aerosol or heating, change may occur in color of a portion of the wrapper enclosing the aerosol generating article. The color sensor may be configured to output a signal corresponding to optical properties (e.g., wavelength of light) corresponding to color of the wrapper, based on light reflected from the wrapper. When the change in color of the portion of the wrapper is detected, the controller **12** may determine that the aerosol generating article inserted into the insertion space has been already used.

[0046] According to an embodiment, the humidity sensor may sense whether the aerosol generating article is excessively moist. For example, the humidity sensor may include the capacitance sensor. The capacitance sensor may include at least one conductor arranged adjacent to the insertion space. The controller **12** may detect whether the aerosol generating article is overly moist, based on a level of the signals corresponding to the permittivity and the like output from the capacitance sensor. For example, the controller **12** may be configured to confirm a range of level including the level of the signal, based on a look-up table, and determine an amount of moisture of the aerosol generating article based on the range of level that has been confirmed.

[0047] According to an embodiment, the cigarette recognition sensor may be configured to sense authenticity and/or type of the aerosol generating article.

[0048] For example, the cigarette recognition sensor may include an optical sensor configured to sense a recognition material (or a recognition marker) on an external surface (e.g., the wrapper) of the aerosol generating article. The optical sensor may irradiate light toward the recognition material (or the recognition marker) of the aerosol generating article, and sense the authenticity and/or the type of the aerosol generating article, based on the light that has been reflected. For example, the recognition material may include a material irradiating light within a specific wavelength band

based on the light that has been irradiated. The controller **12** may detect the authenticity and/or the type of the aerosol generating article, based on a range of the wavelength.

[0049] As another example, the cigarette recognition sensor may include the capacitance sensor. Permittivity in the insertion space may vary according to the type of the aerosol generating article inserted into the insertion space. The controller **12** may be configured to detect the authenticity and/or the type of the aerosol generating article, based on the signals corresponding to the permittivity and the like output from the capacitance sensor.

[0050] As another example, the cigarette recognition sensor may include the inductive sensor. When the conductor is included in the wrapper and/or an internal portion (e.g., the aerosol generating rod) of the aerosol generating article inserted into the insertion portion, properties of the current (e.g., the frequencies, the current values, the voltage values, the inductance, the impedance and the like of the alternating current), which are sensed by the inductive sensor when the aerosol generating article is inserted into the insertion space, may vary according to the type of the aerosol generating article inserted into the insertion space. The controller **12** may detect the authenticity and/or the type of the aerosol generating article that has been inserted, based on the properties of the current output from the inductive sensor or sensed by the inductive sensor.

[0051] The cigarette recognition sensor is not limited to the aforementioned examples, and may also be implemented as various types of sensors configured to sense the authenticity of the aerosol generating article and/or sense the type of the aerosol generating article. In addition, the cigarette recognition sensor may also include arbitrary combinations of the aforementioned examples.

[0052] According to an embodiment, the cartridge sensor may be configured to sense attachment and/or removal of the cartridge. For example, the cartridge sensor may include an inductive sensor, a capacitive sensor, a resistance sensor, a hall sensor (hall integrated circuit (hall IC)), and/or an optical sensor.

[0053] According to an embodiment, the cap sensor may be configured to sense attachment and/or removal of a cap. For example, the cap sensor may include an inductive sensor, a capacitance sensor, a resistance sensor, a touch sensor, a hall IC, and/or an optical sensor. The cap may include a structure covering at least a portion attached to/inserted into the aerosol generating device **1** or covering at least a portion of the housing of the aerosol generating device **1**. When the cap is attached to the housing or removed from the housing, the cap sensor may output a signal corresponding to the attachment or removal, and the controller **12** may sense the attachment or removal of the cap based on the signal corresponding to the attachment or removal.

[0054] According to an embodiment, the motion sensor may be configured to sense motions of the aerosol generating device **1**. The motion sensor may be implemented as at least one of an accelerator sensor or a gyro sensor.

[0055] According to an embodiment, in addition to the aforementioned sensors, the sensor unit **13** may further include at least one of a humidity sensor, an atmospheric pressure sensor, a magnetic sensor, a Global Positioning System (GPS), or a proximity sensor. Functions of the sensors may be intuitively derived by those skilled in the art from names of the sensors, and therefore, detailed descriptions thereof may be omitted.

[0056] According to an embodiment, the output unit **14** may be configured to output information regarding the state of the aerosol generating device **1**. The output unit **14** may include a display, a haptic unit, and/or an acoustic output unit, but is not limited thereto. For example, the information about the aerosol generating device **1** may include a state of charge/discharge of the power source **11** of the aerosol generating device **1**, a state of pre-heating of the heaters (i.e., the heater **18** and the cartridge heater **24**), a state of insertion/removal of the aerosol generating article or the cartridge, a state of attachment and/or removal of the cap, or a state of limitation on use of the aerosol generating device **1** (e.g., when a suspicious object is sensed). The display may be configured to visually provide the user with the information about the state of the aerosol generating device **1**. For example, the display may include a light-emitting diode (LED), a liquid crystal display (LCD)

panel, an organic light-emitting diode (OLED) display panel, and the like. The display including a touch pad may also be used as the input unit **15**. The haptic unit may provide the user with the information about the state of the aerosol generating device **1**, in a haptic manner. For example, the haptic unit may include a vibration motor, a piezoelectric element, an electrical simulation device, and the like. The acoustic output unit may provide the user with the information about the aerosol generating device **1** in an auditory manner. For example, the acoustic output unit may convert an electric signal into an acoustic signal and output the converted signal to the outside.

[0057] According to an embodiment, the power source **11** may be configured to provide power for operations of the aerosol generating device **1**. The power source **11** may include one or more batteries. The power source **11** may be configured to provide power such that the heaters (i.e., the heater **18** and the cartridge heater **24**) may be heated. In addition, the power source **11** may also be configured to provide power for operations of the controller **12**, the sensor unit **13**, the output unit **14**, the input unit **15**, the communication unit **16**, and the memory **17**, i.e., other components included in the aerosol generating device **1**. The power source **11** may include a rechargeable battery or a disposable battery. For example, the power source **11** may include a lithium polymer (LiPoly) battery, but is not limited thereto. The power source **11** may also include a replaceable (removable) battery (hereinafter, referred to as a detachable battery). The detachable battery may be attached to a battery accommodation unit (not shown) provided in the aerosol generating device **1**, or may be removed from the battery accommodation unit. The detachable battery may also be charged in wired and/or wireless manner.

[0058] According to an embodiment, the heaters (i.e., the heater **18** and the cartridge heater **24**) may be configured to receive power from the power source **11** and heat a medium and/or the aerosol generating material in the cartridge. The aerosol generating device **1** may include the heater **18** configured to heat the aerosol generating device **1** and/or the cartridge heater **24** configured to heat the cartridge (i.e., a solid and/or liquid medium).

[0059] According to an embodiment, the heaters (i.e., the heater **18** and the cartridge heater **24**) may include electric resistance heaters. For example, an electric resistance heater may include electric resistance materials such as metals or metal alloys including titanium, zirconium, tantalum, platinum, nickel, cobalt, chromium, hafnium, niobium, molybdenum, tungsten, tin, gallium, manganese, iron, copper, stainless steel, nichrome, and the like. The electric resistance heater may be implemented as a metal wire, a metal plate on which electrically conductive tracks are arranged, a ceramic heating object, and the like.

[0060] According to an embodiment, the heaters (i.e., the heater **18** and the cartridge heater **24**) may include an induction heater. For example, the induction heater may include a susceptor configured to generate heat in response to a magnetic field. In response to an alternating current flowing through the induction coil, the magnetic field may be generated by the induction coil. The magnetic field that has been generated may pass through the heater, and therefore, an eddy current may be generated in the susceptor. The susceptor may be heated based on the generation of the eddy current. According to an embodiment, the susceptor may also be included in (e.g., the aerosol generating rod) the aerosol generating article. In this case, too, the susceptor included in the aerosol generating article may be heated by the induction coil.

[0061] The heaters (i.e., the heater **18** and the cartridge heater **24**) are not limited to the aforementioned examples, and may include various types of heating methods, structures, and components for heating the aerosol generating article and/or the cartridge, or may be replaced by the various types of heating methods, structures, and components.

[0062] According to an embodiment, the input unit **15** may be configured to receive information input from the user. For example, the input unit **15** may include a touch panel, a button, a key pad, a dome switch, a jog wheel, a jog switch, and the like.

[0063] According to an embodiment, as hardware configured to store various type of data processed in the aerosol generating device **1**, the memory **17** may be configured to store data that



has been processed in the controller **12** and data to be processed in the controller **12**. For example, the memory **17** may include at least one type of storage medium from among a flash memory type, a hard disk type, a multimedia card micro type, a card-type memory (e.g., a SD or XD memory), random access memory (RAM), static random access memory (SRAM), read-only memory (ROM), electrically erasable programmable read-only memory (EEPROM), programmable read-only memory (PROM), magnetic memory, magnetic disc, and optical disc. For example, the memory **17** may be configured to store an operation time of the aerosol generating device **1**, a maximum number of puffs, the number of current puffs, at least one temperature profile, data about a smoking pattern of the user, and the like.

[0064] According to an embodiment, the communication unit **16** may include at least one element for communication with another electronic device (e.g., a mobile electronic device). For example, the communication unit **16** may include Bluetooth communication unit, Bluetooth Low Energy (BLE) communication unit, Near Field communication unit, wireless local area network (WLAN) communication unit, Zigbee communication unit, infrared Data Association (IrDA) communication unit, Wireless Fidelity Direct (WFD) communication unit, ultra wideband (UWB) communication unit, Adaptive Network Topology (Ant)+ communication unit, cellular network communication unit, Internet communication unit, computer network (e.g., local area network (LAN) or wide area network (WAN)) communication unit, and the like.

[0065] According to an embodiment, the controller **12** may be configured to control general operations of the aerosol generating device **1**. For example, the controller **12** may include at least one processor. The controller **12** may be implemented as an array of a plurality of logic gates, and may also be implemented as a combination of a general-purpose micro controller unit (MCU) (or a microprocessor) and a memory configured to store programs executable in the MCU. In addition, it will be understood to those skilled in the art that the controller **12** may also be implemented as other types of hardware.

[0066] According to an embodiment, the controller **12** may be configured to control the temperature of the heaters (i.e., the heater **18** and the cartridge heater **24**) by controlling the power of the power source **11** provided to the heaters (i.e., the heater **18** and the cartridge heater **24**). The controller **12** may be configured to control the temperature of the heaters (i.e., the heater **18** and the cartridge heater **24**) and/or the power provided to the heaters (i.e., the heater **18** and the cartridge heater **24**), based on the temperature of the heaters (i.e., the heater **18** and the cartridge heater **24**) sensed by using the temperature sensor (e.g., the sensor unit **13**). The controller **12** may be configured to control the temperature of the heaters (i.e., the heater **18** and the cartridge heater **24**) and/or the power provided to the heaters (i.e., the heater **18** and the cartridge heater **24**), based on the temperature profile and/or a power profile stored in the memory **17**.

[0067] According to an embodiment, the controller **12** may be configured to provide the power (e.g., a voltage and/or a current) provided to the heaters (i.e., the heater **18** and the cartridge heater **24**) by controlling a power conversion circuit (not shown) electrically connected to the heaters (i.e., the heater **18** and the cartridge heater **24**) and the power source **11**. For example, the power conversion circuit may include a direct current (DC)/DC converter (e.g., a buck converter, a buck-boost converter, a boost converter, a Zener diode, and the like) configured to convert power to be provided to the heaters (i.e., the heater **18** and the cartridge heater **24**) and a DC/alternating current (AC) converter (e.g., an inverter) configured to convert power to be provided to the induction coil (not shown). The DC/AC converter may be implemented as a full-bridge circuit or a half-bridge circuit including a plurality of switching elements. For example, the power conversion circuit may include at least one switching element, e.g., a Bipolar Junction Transistor (BJT), a Field Effect Transistor (FET), and the like.

[0068] According to an embodiment, the controller **12** may be configured to adjust a current and/or power provided to the heaters (i.e., the heater **18** and the cartridge heater **24**) by adjusting a frequency and/or a duty ratio of a current pulse input to the at least one switching element of the

power conversion circuit (not shown). The duty ratio of an/off operations of the switching element may correspond to a ratio of an output voltage of the power conversion circuit to an output voltage of the power source **11**.

[0069] According to an embodiment, the controller **12** may be configured to control the power to be provided to the heaters (i.e., the heater **18** and the cartridge heater **24**), by using at least one of a Pulse Width Modulation (PWM) method and a Proportional-Integral-Differential (PID) method. For example, the controller **12** may be configured to control a current pulse having a certain frequency and a certain duty ratio to be provided to the heaters (i.e., the heater **18** and the cartridge heater **24**), by using the PWM method. The controller **12** may be configured to control the power to be provided to the heaters (i.e., the heater **18** and the cartridge heater **24**), by adjusting the frequency and the duty ratio of the current pulse. For example, the controller **12** may be configured to determine a target temperature, which is a target of control, based on the temperature profile. The controller **12** may be configured to control the power provided to the heaters (i.e., the heater **18** and the cartridge heater **24**) by using the PID method. The PID method is a feedback control method performed by using a value of a difference between the temperature of the heaters (i.e., the heater **18** and the cartridge heater **24**) and the target temperature, a value obtained by integrating the value of the difference according to the passage of time, and a value obtained by differentiating the value of the difference according to the passage of time.

[0070] According to an embodiment, the controller **12** may determine target power, which is a target of control, based on the power profile. The controller **12** may also be configured to control the power to be provided to the heaters (i.e., the heater **18** and the cartridge heater **24**) in correspondence to the target power that has been set, according to the passage of time.

[0071] According to an embodiment, the controller **12** may be configured to detect the puffs of the user by sensing the power provided to the heaters (i.e., the heater **18** and the cartridge heater **24**). More particularly, the controller **12** may be configured to control the power to be provided to the heaters (i.e., the heater **18** and the cartridge heater **24**), by using the PID method. When the puffs of the user occur, temperatures may temporally drop in the space into which the aerosol generating article is inserted (hereinafter, referred to as 'insertion space'), the heaters (i.e., the heater **18** and the cartridge heater **24**), and the like. Accordingly, there may be some changes in the power (or the current) provided to the heaters (i.e., the heater **18** and the cartridge heater **24**) while the power is being controlled through the PID method. The controller **12** may be configured to detect the puffs of the user based on the changes in the power under control.

[0072] According to an embodiment, the controller **12** may be configured to prevent the heaters (i.e., the heater **18** and the cartridge heater **24**) from being overheated. For example, the controller **12** may be configured to operations of the power conversion circuit to reduce an amount of the power provided to the heaters (i.e., the heater **18** and the cartridge heater **24**) or stop power supply to the heaters (i.e., the heater **18** and the cartridge heater **24**), based on that the temperature of the heaters (i.e., the heater **18** and the cartridge heater **24**) exceeds a limited temperature that has been already set.

[0073] According to an embodiment, the controller **12** may be configured to control charging and discharging of the power source **11**. For example, the controller **12** may be configured to confirm the temperature of the power source **11** by using the temperature sensor (e.g., the sensor unit **13**). When the temperature of the power source **11** is a first limited temperature or higher, the controller **12** may interrupt charging of the power source **11**. When the temperature of the power source **11** is a second limited temperature or higher, the controller **12** may stop the use (e.g., discharge) of the power stored in the power source **11**. The controller **12** may be configured to calculate a remaining amount of the power stored in the power source **11**. For example, the controller **12** may be configured to calculate a remaining amount of the power source **11**, based on a value obtained by sensing a voltage and/or a current of the power source **11**.

[0074] According to an embodiment, the controller **12** may be configured to control the power

provided to the heaters (i.e., the heater **18** and the cartridge heater **24**), based on a result sensed by the sensor unit **13**.

[0075] According to an embodiment, the controller **12** may be configured to control the power supply to the heaters (i.e., the heater **18** and the cartridge heater **24**), based on the insertion and/or removal of the aerosol generating article into/from the insertion space. For example, when it is determined that the aerosol generating article is inserted into the insertion space, by using the insertion sensor (e.g., the sensor unit **13**), the controller **12** may control the power to be supplied to the heaters (i.e., the heater **18** and the cartridge heater **24**). When it is determined that the aerosol generating article is removed from the insertion space, by using the insertion sensor (e.g., the sensor unit **13**), the controller **12** may interrupt the power supply to the heaters (i.e., the heater **18** and the cartridge heater **24**). When the temperature of the heaters (i.e., the heater **18** and the cartridge heater **24**) is the limited temperature or higher, or when a gradient of the change in the temperature of the heaters (i.e., the heater **18** and the cartridge heater **24**) is a set gradient or greater, the controller **12** may also determine that the aerosol generating article has been removed from the insertion space.

[0076] According to an embodiment, the controller **12** may be configured to control a time period and/or an amount of power supply to the heaters **18** and **24**, based on the state of the aerosol generating article. For example, when the aerosol generating article is determined as overly moist, by using the humidity sensor (e.g., the sensor unit **13**), the controller **12** may increase the time period of power supply (e.g., a pre-heating time period) to the heaters (i.e., the heater **18** and the cartridge heater **24**).

[0077] According to an embodiment, the controller **12** may be configured to control the power supply to the heaters (i.e., the heater **18** and the cartridge heater **24**), based on whether the aerosol generating article has been reused. For example, when it is determined that the aerosol generating article has been already used, the controller **12** may interrupt the power supply to the heaters (i.e., the heater **18** and the cartridge heater **24**).

[0078] According to an embodiment, the controller **12** may control the power supply to the heaters (i.e., the heater **18** and the cartridge heater **24**), based on whether the cartridge has been combined to/removed from (the aerosol generating device **1**). For example, when it is determined that the cartridge has been removed from the aerosol generating device **1**, by using the cartridge sensor (e.g., the sensor unit **13**), the controller **12** may stop the power supply to the heaters (i.e., the heater **18** and the cartridge heater **24**), or may control the power to be not supplied to the heaters (i.e., the heater **18** and the cartridge heater **24**).

[0079] According to an embodiment, the controller **12** may control the power supply to the heaters (i.e., the heater **18** and the cartridge heater **24**), based on whether the aerosol generating material in the cartridge has been used up. For example, when it is determined that the temperature of the heaters (i.e., the heater **18** and the cartridge heater **24**) exceeds the limited temperature while the heaters **18** and **24** are being pre-heated (i.e., in a pre-heating section), the controller **12** may determine that the aerosol generating material in the cartridge has been used up. When it is determined that the aerosol generating material in the cartridge has been used up, the controller **12** may interrupt the power supply to the heaters (i.e., the heater **18** and the cartridge heater **24**).

[0080] According to an embodiment, the controller **12** may control the power supply to the heaters (i.e., the heater **18** and the cartridge heater **24**), based on whether the cartridge is available. For example, when it is determined that the number of current puffs is determined as a maximum number of puffs or set for the cartridge or greater, based on the data stored in the memory **17**, the controller **12** may determine that the cartridge is unavailable. Alternatively, when a total time period of heating the heaters (i.e., the heater **18** and the cartridge heater **24**) is a preset maximum time period or longer, or a total amount of power supplied to the heaters (i.e., the heater **18** and the cartridge heater **24**) is a preset maximum amount of power or greater, the controller **12** may determine that the cartridge is unavailable. In this case, the controller **12** may stop the power supply

to the heaters (i.e., the heater **18** and the cartridge heater **24**) or may control the power to be not provided to the heaters (i.e., the heater **18** and the cartridge heater **24**).

[0081] According to an embodiment, the controller **12** may control the power supply to the heaters (i.e., the heater **18** and the cartridge heater **24**), based on the puffs of the user. For example, the controller **12** may determine the occurrence of the puffs and/or the intensity of the puffs, by using the puff sensor (e.g., the sensor unit **13**). When the number of puffs reaches a preset maximum number of puffs and/or the puffs are not sensed for a preset time period or longer, the controller **12** may interrupt the power supply to the heaters (i.e., the heater **18** and the cartridge heater **24**). When the puffs are sensed, the controller **12** may also control the power supply to the heaters (i.e., the heater **18** and the cartridge heater **24**).

[0082] According to an embodiment, the controller **12** may control the power supply to the heaters (i.e., the heater **18** and the cartridge heater **24**), based on the authenticity and/or the type of the aerosol generating article (or the cartridge). For example, the controller **12** may detect the authenticity and/or the type of the aerosol generating article by using the cigarette recognition sensor (e.g., the sensor unit **13**). For example, when it is determined that the aerosol generating article (or the cartridge) is inauthentic, the controller **12** may interrupt the power supply to the heaters (i.e., the heater **18** and the cartridge heater **24**). When it is detected that the aerosol generating article (or the cartridge) is authentic, the controller **12** may control (e.g., initiate) the power supply to the heaters (i.e., the heater **18** and the cartridge heater **24**). As another example, the controller **12** may differently control power supply to the heaters (i.e., the heater **18** and the cartridge heater **24**) according to the type of the aerosol generating article (or the cartridge). More particularly, when it is detected that the aerosol generating article (or the cartridge) is a first aerosol generating article (or a first cartridge), the controller **12** may control the temperature and/or the power of the heaters (i.e., the heater **18** and the cartridge heater **24**) based on a first temperature profile (or a first power profile), and when it is detected that the aerosol generating article (or the cartridge) is a second aerosol generating article (or a second cartridge), the controller **12** may control the temperature and/or the power of the heaters (i.e., the heater **18** and the cartridge heater **24**) based on a second temperature profile (or a second power profile).

[0083] According to an embodiment, the controller **12** may control the output unit **14**, based on a result detected by the sensor unit **13**. For example, when the number of puffs counted by using the puff sensor (e.g., the sensor unit **13**) reaches a preset number, the controller **12** may control the output unit **14** to provide information indicating that the operation of the aerosol generating device **1** will be soon finished, in a visual manner, a haptic manner, and/or an auditory manner. For example, the controller **12** may control the output unit **14** to provide information about the temperature of the heaters (i.e., the heater **18** and the cartridge heater **24**) in a visual manner, a haptic manner, and/or an auditory manner.

[0084] According to an embodiment, the controller **12** may, based on occurrence of a certain event, store and update history about the event in the memory **17**. For example, the event may include sensing of insertion of the aerosol generating device **1**, initiation of heating of the aerosol generating material, sensing of the puffs, end of the puffs, sensing overheating of the heaters (i.e., the heater **18** and the cartridge heater **24**), sensing application of an overvoltage to the heaters (i.e., the heater **18** and the cartridge heater **24**), end of heating of the aerosol generating article, operations such as power on/off of the aerosol generating device **1**, initiation of charging the power source **11**, sensing overcharging of the power source **11**, end of charging of the power source **11**, and the like. For example, the history of the event may include a date on which the event occurred, log data corresponding to the event, and the like. For example, when the certain event is sensing insertion of the aerosol generating article, the log data corresponding to the event may include data about a sensing value of the insertion sensor (e.g., the sensor unit **13**) and the like. For example, when the certain event is sensing that the heaters (i.e., the heater **18** and the cartridge heater **24**) are overheated, the log data corresponding to the event may include data about the temperature of the

heaters (i.e., the heater **18** and the cartridge heater **24**), the voltage applied to the heaters (i.e., the heater **18** and the cartridge heater **24**), a current flowing through the heaters (i.e., the heater **18** and the cartridge heater **24**) and the like.

[0085] According to an embodiment, the controller **12** may control the communication unit **16** to form a communication link with an external device such as a mobile terminal of the user.

[0086] According to an embodiment, when data about authentication is received from the external device through the communication link, the controller **12** may release limitation on use of at least one function (e.g., a heating function) of the aerosol generating device **1**. For example, the data about authentication may include birthday of the user, an identification number indicating the user, whether the user has been authenticated, and the like.

[0087] According to an embodiment, the controller **12** may transmit data about the state of the aerosol generating device **1** (e.g., a remaining capacity of the power source **11**, operation modes, and the like) to the external device through the communication link. The data that has been transmitted may be output through a display of the external device and the like.

[0088] According to an embodiment, when a request to search for the location of the aerosol generating device **1** is received from the external device through the communication link, the controller **12** may control the output unit **14** to perform an operation corresponding to searching for the location of the aerosol generating device **1**. For example, the controller **12** may control the haptic unit to cause vibration or control the display to output an object corresponding to searching for the location and end of the searching for the location.

[0089] According to an embodiment, when firmware data is received from the external device through the communication link, the controller **12** may perform firmware update.

[0090] According to an embodiment, the controller **12** may transmit data about the sensing value of at least one sensor unit **13** to an external server (not shown) through the communication link, and may receive and store, from the external server, a learning model generated by learning the sensing value through machine learning such as deep learning. The controller **12** may perform an operation of determining a puff pattern of the user, an operation of generating temperature profiles, and the like, by using the learning model received from the external server.

[0091] Although not shown in FIG. **1**, the aerosol generating device **1** may further include a power protection circuit. The power protection circuit may include at least one switching element, and may block an electric line for the power source **11** in correspondence to overcharging and/or over-discharging of the power source **11**. The aerosol generating device **1** may further include a connection interface such as Universal Serial Bus (USB) interface, and may transmit/receive information or charge the power source **11** by being in connection with another external device through the connection interface.

[0092] The aerosol generating article mentioned in the disclosure may include at least one aerosol generating rod and at least one filter rod. The heater **18** may be arranged in correspondence to the at least one aerosol generating rod, and may be differently designed according to arrangement orders and/or locations of the aerosol generating rod and the filter rod. The aerosol generating rod may include at least one of nicotine, the aerosol generating material, and additives. For example, the aerosol generating material may include glycerin (e.g., vegetable glycerin (VG) and/or propylene glycol (PG)), and may further include other various materials. For example, the additives may include flavoring agent and/or organic acids, and may further include other various materials. For example, the aerosol generating rod may include an aerosol generating substrate (e.g., a sheet) into which a non-tobacco material having a liquid phase (e.g., the aerosol generating material and/or nicotine) is impregnated, and/or may include a tobacco material having a solid phase (e.g., leaf tobacco, reconstituted tobacco, and the like). The tobacco material may be included in the aerosol generating rod, in various forms such as cut tobacco, granules, and powders. According to an embodiment, the additives in the aerosol generating rod may include alkaline materials. Based on the alkaline materials, nicotine in the tobacco material included in the aerosol generating rod

may have an alkaline pH (e.g., pH 7.0 or higher). In this case, even under a low temperature, freebase nicotine may be released from the aerosol generating rod. According to an embodiment, the aerosol generating rod may include two or more aerosol generating rods, and the two or more aerosol generating rods may each include tobacco materials and/or non-tobacco materials. Although not shown, the at least one aerosol generating rod and the at least one filter rod may each and/or integrally be covered with at least one wrapper. In the disclosure, the aerosol generating article may also be referred to as a stick.

[0093] The cartridge mentioned in the disclosure may contain an aerosol generating material in any one of a liquid state, a solid state, a gas state, or a gel state. The aerosol generating material may include a liquid composition. For example, the liquid composition may include liquid containing a tobacco-containing material including volatile tobacco-flavor components, and may also include liquid containing a non-tobacco material. The cartridge may include a storage including the aerosol generating material and/or a liquid delivery element containing (or impregnated with) the aerosol generating material. For example, the liquid delivery element may include a wick such as cotton fibers, ceramic fibers, glass fibers, and porous ceramic. The cartridge heater **24** may be included in the cartridge, in a coil-type structure surrounding (or winding) the liquid delivery element or a structure being in contact with a side of the liquid delivery element. Alternatively, the cartridge heater **24** may also be included in the aerosol generating device **1** that may be separated from the cartridge.

[0094] FIG. **2** illustrates the aerosol generating device **1** according to an embodiment. FIG. **3** illustrates the aerosol generating device **1** according to an embodiment.

[0095] According to an embodiment, the aerosol generating device **1** may include a housing **10**, the power source **11**, the controller **12**, the sensor unit **13**, and/or heaters **182** and **183** (e.g., the heater **18** in FIG. **1**). However, it will be understood to those skilled in the art that elements included in the aerosol generating device **1** are not limited to the elements shown in FIG. **2** or FIG. **3**, and some of the elements may be omitted or other elements may be added. The aerosol generating device **1** shown in FIG. **2** may be referred to as an ‘internal-heating type’ aerosol generating device configured to heat an internal portion of the aerosol generating article **2**. The aerosol generating device **1** shown in FIG. **3** may be referred to as an ‘external-heating type’ aerosol generating device configured to heat an external portion of the aerosol generating article **2**. Regarding the following drawings, descriptions with reference to FIG. **1** will be omitted.

[0096] According to an embodiment, the housing **10** may provide a space that is open upward for insertion of the aerosol generating article **2**. In the disclosure, the space open upward may be referred to as the insertion space. The insertion space may be formed by being recessed to a certain depth toward an internal portion of the housing **10** such that at least a portion of the aerosol generating article **2** may be inserted into the housing **10**. A depth of the insertion space may be equal to or greater than a length of an area including the aerosol generating material and/or a medium in the aerosol generating article **2**. A lower end of the aerosol generating article **2** may be inserted into the housing **10**, and an upper end of the aerosol generating article may protrude outside the housing **10**. The user may hold the upper end of the aerosol generating article **2** exposed outside, in the mouth, and puff the aerosol.

[0097] According to an embodiment, the heaters **182** and **183** may heat the aerosol generating article **2**.

[0098] Referring to FIG. **2**, the heater **182** may include an internal-heating type heater.

[0099] According to an embodiment, the internal-heating type heater may extend upward to a great length in the space into which the aerosol generating article **2** inserted (i.e., the insertion space). For example, the internal-heating type heater may include a heating element having the form of a rod or a pin, as illustrated, but may also include various heating elements, e.g., a tube-type heating element or a plate-type heating element. The internal-heating type heater may be inserted through a lower portion of the aerosol generating article **2**.

[0100] According to an embodiment, the internal-heating type heater may include an electric resistance heater or an induction heater.

[0101] For example, the electric resistance heater may include an electric resistance material inside (e.g., an internal hole or an inner surface) or outside (e.g., an outer surface), and may be heated in response to a current flowing through the electric resistance material. In this case, the electric resistance heater may be electrically connected to the power source **11**, and may receive the current from the power source **11** and directly generate heat. In addition, the induction coil **181** may also be omitted.

[0102] For example, in a case of the induction-heating type heater, the aerosol generating device **1** may include the induction coil **181** surrounding at least a portion of the internal-heating type heater (e.g., arranged outside to correspond to a length of the at least a portion of the internal-heating type heater). In this case, a magnetic flux concentrator and the like may be further included outside the induction coil **181** to improve the efficiency of induction heating. The induction heater may include a susceptor, and may generate heat based on the magnetic field generated from the induction coil **181**. According to an embodiment, the induction heater (e.g., the susceptor) (or a heater module including the induction heater) may be arranged to be detachable from the housing **10**.

[0103] According to an embodiment, the heater **182** may include multiple heaters. The multiple heaters may include a first heater and a second heater, and may be inserted into the aerosol generating article **2**. The first heater and the second heater may be arranged in parallel in a longitudinal direction. The first heater and the second heater may operate as the electric resistance heater and/or the induction heater, may be sequentially heated, or may be simultaneously heated. In this case, the first heater and the second heater may also be respectively arranged at locations corresponding to locations in the longitudinal direction of the two or more aerosol generating rods. Alternatively, the first heater and the second heater may also be respectively arranged at locations corresponding to locations in a longitudinal direction of a first portion and a second portion of an aerosol generating rod. When the heater **182** is an induction heater, the aerosol generating device **1** may include a first induction coil and a second induction coil, and the first induction coil and the second induction coil may also be respectively arranged at locations corresponding to locations in a longitudinal direction of the first heater and the second heater. Alternatively, the first heater and the second heater may also be respectively arranged at locations corresponding to locations in a longitudinal direction of a first portion and a second portion of the heater **182**. In addition, three or more heaters and/or induction coils may be included.

[0104] According to an embodiment, the disclosure may also be implemented such that the susceptor is arranged (or included) in the aerosol generating article **2** (e.g., a medium unit) and the susceptor included in the aerosol generating article **2** generates heat based on the magnetic field generated from the induction coil **181**.

[0105] Referring to FIG. **3**, the heater **183** may include an external-heating type heater.

[0106] According to an embodiment, the external-heating type heater may extend upward to a great length around the space into which the aerosol generating article **2** is inserted (i.e., the insertion space). For example, the external-heating type heater may be arranged to surround at least a portion of the insertion space. For example, the external-heating type heater may include a tube type heater (e.g., a cylinder shape) including a hole therein. The external-heating type heater may also include the form including a hole therein and surrounding the hole. In this case, the external-heating type heater may be supported by a polyimide film. The heater supported by such a film may be referred to as a film heater. The external-heating type heater may be arranged to surround at least a portion of the insertion space. The external-heating type heater may heat the outside of the aerosol generating article **2** inserted into the hole.

[0107] According to an embodiment, the external heater may include an electric resistance heater or an induction heater, and same descriptions as those with reference to FIG. **2** will be omitted. In the induction heater, the aerosol generating device **1** may include an external heater implemented as

a susceptor having the form of a tube, and may include the induction coil **181** surrounding at least a portion of the external heater (e.g., arranged outside to correspond to a length of at least a portion of the external heater). In addition, the induction coil **181** may also include a fan coil. When the external-heating type heater includes an electric resistance heater, heat may be generated through a current flow in the electric resistance heater (e.g., the film heater) having the form of a tube, and therefore, addition of the induction coil **181** may be omitted. An insulator may be arranged outside the external-heating type heater. By doing so, heat dissipated in a radially outward direction from the heater **183** and applied to the outside of the housing **10** may be reduced.

[0108] According to an embodiment, the heater **183** may include multiple heaters, and a first heater and a second heater may be arranged in parallel in the longitudinal direction, and may each surround at least a portion of the insertion portion. The first heater and the second heater may operate as the electric resistance heater and/or the induction heater, may be sequentially heated, or may be simultaneously heated. When the heater is an induction heater, the aerosol generating device **1** may include a first induction coil and a second induction coil, and the first induction coil and the second induction coil may also be respectively arranged at locations corresponding to locations in the longitudinal direction of the first heater and the second heater. Alternatively, the first heater and the second heater may also be respectively arranged at locations corresponding to locations in a longitudinal direction of a first portion and a second portion of the heater **183**.

[0109] Unlike in FIG. **2** or FIG. **3**, the heater **182** in FIG. **2** and the heater **183** in FIG. **3** may be included together in the aerosol generating device **1**. In this case, the heater **182** may heat the inside of the aerosol generating article **2**, and the heater **183** may heat the outside of the aerosol generating article **2**.

[0110] According to an embodiment, an airflow channel through which air flows may be provided in the aerosol generating device **1**. For example, the housing **10** may include a structure (e.g., a hole) through which the air may be introduced into the housing **10** from the outside. The air introduced into the housing **10** may be introduced into the aerosol generating article through the lower end (that is, an upstream side) of the aerosol generating article **2**. Together with the air that has been introduced, the aerosol generated based on the heating of the aerosol generating article **2** may be puffed into an oral cavity of the user through the upper end (that is, a downstream side) of the aerosol generating article **2**.

[0111] FIG. **4** illustrates the aerosol generating device **1** according to an embodiment.

[0112] According to an embodiment, the aerosol generating device **1** may include the housing **10**, the power source **11**, the controller **12**, the sensor unit **13**, and/or the heaters **183** and **24** (i.e., the heater **183** and the cartridge heater **24**) (e.g., the heaters (i.e., the heater **18** and the cartridge heater **24**) in FIG. **1**). However, it will be understood to those skilled in the art that elements included in the aerosol generating device **1** are not limited to the elements shown in FIG. **4**, and some of the elements may be omitted or other elements may be added. Regarding the following drawings, descriptions with reference to FIG. **1** will be omitted.

[0113] According to an embodiment, the housing **10** may provide the space open upward for insertion of the aerosol generating article **2** (hereinafter, will be referred to as 'insertion space'). The insertion space may be formed by being recessed to a certain depth toward an internal portion of the housing **10** such that at least a portion of the aerosol generating article **2** may be inserted into the housing **10**. The lower end of the aerosol generating article **2** may be inserted into the housing, and the upper end of the aerosol generating article **2** may protrude outside the housing **10**.

[0114] Unlike in the drawings, a cartridge **19** may provide the insertion space for accommodating the aerosol generating article **2**. In this case, the insertion space may be formed by being recessed to a certain depth toward an inner portion of the cartridge **19** such that at least a portion of the aerosol generating article **2** may be inserted into the cartridge **19**. The lower end of the aerosol generating article **2** may be inserted into the cartridge **19**, and the upper end of the aerosol generating article **2** may protrude outside the cartridge **19**. In addition, in this case, the aerosol



generating device **1** may not include the heater **183**.

[0115] According to an embodiment, a depth of the insertion space may be equal to or greater than the length of the area including the aerosol generating material and/or a medium in the aerosol generating article **2**. The user may hold the upper end of the aerosol generating article **2** exposed outside, by the mouth, and inhale the air.

[0116] According to an embodiment, the heater **183** may heat the aerosol generating article **2**. The heater **183** may extend upward to a great length around the space into which the aerosol generating article **2** is inserted (i.e., the insertion space). For example, the heater **183** may have the form of a tube (e.g., a cylinder shape) including a hole therein. The heater **183** may include the form including a hole inside and surrounding the hole. In this case, the heater **183** may be supported by a polyimide film. The heater supported by such a film may also be referred to as a film heater. The heater **183** may be arranged to surround at least a portion of the insertion space. The heater **183** may heat the outside of the aerosol generating article **2** inserted into the hole. In the disclosure, the heater **183** may be referred to an external heater heating the outside of the aerosol generating article. An insulator may be arranged outside the heater **183**. By doing so, heat dissipated in the radially outward direction from the heater **183** and applied to the outside of the housing **10** may be reduced.

[0117] According to an embodiment, the heater **183** may include an electric resistance heater and/or an induction heater.

[0118] For example, the electrical resistance heater may include an electrical resistance material, and may be heated in response to a current flowing through the electrical resistance material. In this case, the electric resistance heater may be electrically connected to the power source **11**, and may receive the current from the power source **11** and directly generate heat.

[0119] For example, in a case of an induction heater, the aerosol generating device **1** may further include an induction coil (not shown) surrounding at least a portion of the heater **183** (e.g., arranged outside the heater **183** in correspondence to a length of the at least portion of the heater **183**). In this case, a magnetic flux concentrator and the like to improve the efficiency of induction heating may be further included outside the induction coil (not shown). The induction heater may include a susceptor, and may generate heat based on a magnetic field generated from the induction coil (not shown).

[0120] According to an embodiment, the heater **183** may include the multiple heaters. The multiple heaters may include the first heater and the second heater, and may be inserted into the aerosol generating article **2**. The first heater and the second heater may be arranged in parallel in a longitudinal direction. The first heater and the second heater may operate as the electric resistance heater and/or the induction heater, may be sequentially heated, or may be simultaneously heated. In this case, the first heater and the second heater may also be respectively arranged at the locations corresponding to the locations in the longitudinal direction of the two or more aerosol generating rods. Alternatively, the first heater and the second heater may also be respectively arranged at the locations corresponding to locations in a longitudinal direction of a first portion and a second portion of an aerosol generating rod. When the heater is an induction heater, the aerosol generating device **1** may include the first induction coil and the second induction coil, and the first induction coil and the second induction coil may also be respectively arranged at locations corresponding to the locations in the longitudinal direction of the first heater and the second heater. Alternatively, the first heater and the second heater may also be respectively arranged at locations corresponding to locations in a longitudinal direction of a first portion and a second portion of the heater **183**. In addition, three or more heaters and/or induction coils may be included.

[0121] Unlike the drawing, the aerosol generating device **1** may not include the heater **183**. The aerosol generating article **2** may be directly or indirectly heated by the cartridge heater **24**, or may be substantially not heated. Indirect heating may indicate that the aerosol generating article **2** receives heat accommodated in the aerosol and is heated in a process where the aerosol generated

by the cartridge heater **24** passes through the aerosol generating article **2**. In this case, the aerosol generating device **1** may be referred to as a non-heating (or an indirect heating) aerosol generating device. The aerosol generating rod of the aerosol generating article **2** may include additives such as alkaline materials. Based on the alkaline materials, nicotine included in the aerosol generating rod may have an alkaline pH (e.g., pH 7.0 or higher). Such alkaline nicotine may flow into the oral cavity of the user together with the aerosol introduced into the aerosol generating article **2** from the cartridge **19** described below.

[0122] Unlike in the drawings, the heater **183** may also include an internal heater. For example, the internal heater may include various types of heating elements, e.g., a rod-type heating element, a tube-type heating element, a plate-type heating element, a needle-type heating element, or the like. The internal heater may be inserted through the lower portion of the aerosol generating article **2**, and may be set to heat the inside of the aerosol generating article **2**.

[0123] According to an embodiment, the cartridge **19** may be combined to the housing **10** in a detachable manner. For example, as a space may be formed at a side of the housing **10** and at least a portion of the cartridge **19** is inserted into the space formed at the side of the housing **10**, the cartridge **19** may be mounted in the housing **10**. Alternatively, the cartridge **19** may be formed integral with the housing **10**.

[0124] According to an embodiment, an airflow channel through which air flows may be provided in the aerosol generating device **1** and/or the cartridge **19**. For example, in a state where the cartridge **19** has been inserted, the housing **10** may include a structure in which air may be introduced into the housing **10** from the outside. The air that has been introduced into the housing may pass through the cartridge **19** and may be introduced into the insertion space through an airflow channel CN, and may flow into the oral cavity of the user. The airflow channel CN may also include various structures for reducing a remaining amount of droplets or facilitating flow of the air.

[0125] Although FIG. **4** illustrates that the cartridge **19** is located laterally to the aerosol generating article **2** and the airflow channel CN is formed from a lateral position of the aerosol generating article **2** to the lower end (that is, the upstream side) of the aerosol generating article **2**, locations of the cartridge **19** and the airflow channel CN are not limited thereto. For example, the cartridge **19** may be located adjacent to the lower end (that is, the upstream side) of the aerosol generating article **2**, and in this case, the airflow channel CN may be substantially formed in a straight-line shape to connect the cartridge **19** to the lower end (that is, the upstream side) of the aerosol generating article **2**.

[0126] According to an embodiment, the cartridge **19** may include a chamber CO containing the aerosol generating material, the cartridge heater **24**, and/or a liquid delivery element impregnated with (or containing) the aerosol generating material. The liquid delivery element **25** may be impregnated with the aerosol generating material provided from the chamber CO. For example, the liquid delivery element may include a wick such as cotton fiber, ceramic fiber, glass fiber, and porous ceramic.

[0127] According to an embodiment, the cartridge heater **24** may heat the aerosol generating material contained in the cartridge **19**. For example, the cartridge heater **24** may include an electrical resistance heater and/or an induction heater.

[0128] For example, the electrical resistance heater may include an electrical resistance material, and may be heated as a current flows through the electrical resistance material. As another example, in a case of the induction heater, the aerosol generating device **1** may further include an induction coil (not shown) around the induction heater. The induction heater may include a susceptor, and may generate heat based on a magnetic field generated from the induction coil (not shown). The cartridge heater **24** may be formed in the form of a coil surrounding (or winding) the liquid delivery element and/or the form of being in contact with a side of the liquid delivery element.

[0129] Unlike in the drawing, the cartridge heater **24** may also be included in the aerosol generating device **1**. For example, the cartridge heater **24** may be included in the housing **10**. In this case, by removing the cartridge **19**, the cartridge **19** and the cartridge heater **24** may be separated from each other.

[0130] According to an embodiment, the aerosol may be generated based on the heating of the cartridge heater **24**. For example, as the aerosol generating material impregnated into the liquid delivery element is heated by the cartridge heater **24**, vapor may be generated from the aerosol generating material, and as the vapor that has been generated is mixed with an external air introduced into the cartridge **19**, the aerosol may be generated. The aerosol generated by the cartridge heater **24** may be introduced into the aerosol generating article through the airflow channel CN. While the aerosol passes through the aerosol generating article **2**, tobacco or a flavoring material may be added to the aerosol, and the aerosol, to which the tobacco or the flavoring material has been added, may be inhaled into the oral cavity of the user through an end of the aerosol generating article **2**.

[0131] FIGS. **5** and **6** are diagrams respectively illustrating examples of the aerosol generating article **2**.

[0132] Referring to FIG. **5**, the aerosol generating article **2** includes an aerosol generating rod **21** and a filter rod **22**.

[0133] FIG. **5** illustrates that the filter rod **22** includes a single segment, but is limited thereto. In other words, the filter rod **22** may include a plurality of segments. For example, the filter rod **22** may include a first segment configured to cool an aerosol and a second segment configured to filter a certain component included in the aerosol. Also, as necessary, the filter rod **22** may further include at least one segment configured to perform other functions.

[0134] The aerosol generating article **2** may have a diameter in a range from about 5 mm to about 9 mm and a length about 48 mm, but is not limited thereto. For example, a length of the aerosol generating rod **21** may be about 12 mm, a length of a first segment of the filter rod **22** may be about 10 mm, a length of a second segment of the filter rod **22** may be about 14 mm, and a length of a third segment of the filter rod **22** may be about 12 mm, but the embodiment is not limited thereto.

[0135] The aerosol generating article **2** may be packaged by at least one wrapper **24W**. The wrapper **24W** may have at least one hole through which external air may be introduced or internal air may be discharged. For example, the aerosol generating article **2** may be packaged by one wrapper **24W**. As another example, the aerosol generating article **2** may be doubly packaged by two or more wrappers **24W**. For example, the aerosol generating rod **21** may be package by a first wrapper **241**. For example, the filter rod **22** may be packaged by a plurality of individual wrappers (i.e., a second wrapper **242**, a third wrapper **243**, and a fourth wrapper **244**). The aerosol generating rod **21** and the filter rod **22**, which have been packaged by the individual wrappers, may be combined to each other. The entire portion of the aerosol generating article **2** may be packaged again by a fifth wrapper **245**. When the filter rod **22** includes a plurality of segments, the plurality of segments may be respectively packaged by the plurality of individual wrappers (i.e., the second wrapper **242**, the third wrapper **243**, and the fourth wrapper **244**). The entire portion of the aerosol generating article **2**, in which the segments packaged by the individual wrappers are combined to one another, may be packaged again by another wrapper.

[0136] The first wrapper and the second wrapper **242** may be manufactured with standard filter rolling papers. For example, the first wrapper **241** and the second wrapper **242** may include porous rolling paper or non-porous rolling paper. In addition, the first wrapper **241** and the second wrapper **242** may also be manufactured with oil-resistant paper and/or an aluminum-laminated packaging material.

[0137] The third wrapper **243** may be manufactured with hard rolling paper. For example, a basis weight of the third wrapper **243** may be in a range from about 88 g/m.sup.2 to about 96 g/m.sup.2. For example, the basis weight of the third wrapper **243** may be in a range from about 90 g/m.sup.2

to about 94 g/m.<sup>2</sup>. In addition, a thickness of the third wrapper **243** may be in a range from about 120  $\mu\text{m}$  to about 130  $\mu\text{m}$ . For example, the thickness of the third wrapper **243** may be about 125  $\mu\text{m}$ .

[0138] The fourth wrapper **244** may be manufactured with oil-resistant hard rolling paper. For example, a basis weight of the fourth wrapper **244** may be in a range from about 88 g/m.<sup>2</sup> to about 96 g/m.<sup>2</sup>. For example, the basis weight of the fourth wrapper **244** may be in a range from about 90 g/m.<sup>2</sup> to about 94 g/m.<sup>2</sup>. In addition, a thickness of the fourth wrapper **244** may be in a range from about 120  $\mu\text{m}$  to about 130  $\mu\text{m}$ . For example, the thickness of the fourth wrapper **244** may be about 125  $\mu\text{m}$ .

[0139] The fifth wrapper **245** may be manufactured with sterile paper (MFW). Here, the sterile paper (MFW) may indicate paper fabricated in a particular manner such that a tensile strength, water resistance, smoothness thereof are improved compared with general paper. For example, a basis weight of the fifth wrapper **245** may be in a range from about 57 g/m.<sup>2</sup> to about 63 g/m.<sup>2</sup>. For example, the basis weight of the fifth wrapper **245** may be about 60 g/m.<sup>2</sup>. In addition, a thickness of the fifth wrapper **245** may be in a range from about 64  $\mu\text{m}$  to about 70  $\mu\text{m}$ . For example, the thickness of the fifth wrapper **245** may be about 67  $\mu\text{m}$ .

[0140] Certain materials may be added to the fifth wrapper **245**. Here, silicon may be an example of the certain materials, but the embodiment is not limited thereto. For example, the silicon may have properties such as heat resistance, i.e., the property of not being easily deformed due to temperatures, oxidation resistance, i.e., the property of not being easily oxidized, resistance against various types of drugs, water-repellency, electrical resistivity, or the like. Although, any material having the aforementioned properties, even the material is not silicon, may be applied or coated onto the fifth wrapper **245** without limitation.

[0141] The fifth wrapper **245** may prevent combustion of the aerosol generating article **2**. For example, when the aerosol generating rod **21** is heated by the heater **110**, it is likely that the aerosol generating article **2** will be combusted. More particularly, when the temperature increases up to or beyond an ignition point of any one of the materials included in the aerosol generating rod **21**, the aerosol generating article **2** may be combusted. Even in this case, the wrapper **245** includes a non-combustible material, the combustion of the aerosol generating article **2** may be prevented.

[0142] In addition, the fifth wrapper **245** may prevent contamination of the aerosol generating device **1** due to the materials generated in the aerosol generating article **2**. By puffs of the user, liquid materials may be generated in the aerosol generating article **2**. For example, as aerosol generated in the aerosol generating article **2** is cooled by external air, the liquid materials (e.g., moisture) may be generated. As the fifth wrapper **245** packages the aerosol generating article **2**, leakage of the liquid materials generated in the aerosol generating article **2** to the outside of the aerosol generating article **2** may be prevented.

[0143] The aerosol generating rod **21** may include an aerosol generating material. For example, the aerosol generating material may include at least one of glycerin, propylene glycol, ethylene glycol, dipropylene glycol, diethylene glycol, triethylene glycol, tetraethylene glycol, and oleyl alcohol, but it is not limited thereto. Also, the aerosol generating rod **21** may include other additives, such as flavors, a wetting agent, and/or organic acid. Also, the aerosol generating rod **21** may include a flavored liquid, such as menthol or a moisturizer, which is injected to the aerosol generating rod **21**.

[0144] The aerosol generating rod **21** may be manufactured in various forms. For example, the aerosol generating rod **21** may be formed as a sheet. For example, the aerosol generating rod **21** may be formed as a strand. For example, the aerosol generating rod **21** may be formed as a pipe tobacco, which is formed of tiny bits cut from a tobacco sheet. For example, the aerosol generating rod **21** may be surrounded by a heat conductive material. For example, the heat conductive material may be, but is not limited to, a metal foil such as aluminum foil. For example, the heat conductive material surrounding the aerosol generating rod **21** may uniformly distribute heat transmitted to the aerosol generating rod **21**, and thus, the heat conductivity applied to the tobacco rod may be

increased. This can improve the taste of the cigarette. Also, the heat conductive material surrounding the aerosol generating rod **21** may function as a susceptor heated by the induction heater. Here, although not illustrated in the drawings, the aerosol generating rod **21** may further include an additional susceptor, in addition to the heat conductive material surrounding the aerosol generating rod **21**.

[0145] The filter rod **22** may include a cellulose acetate filter. Shapes of the filter rod **22** are not limited. For example, the filter rod **22** may include a cylinder-type. For example, the filter rod **22** may include a tube-type rod having a hollow inside. Also, the filter rod **22** may include a recess-type rod. When the filter rod **22** includes a plurality of segments, at least one of the plurality of segments may have a different shape.

[0146] A first segment **221** of the filter rod **22** may include a cellulose acetate filter. For example, the first segment **221** may include a structure having the form of a tube including hole therein. By the first segment **221**, it is possible to prevent a situation that internal materials of the aerosol generating rod **21** are pushed backward when the heater **182** is inserted, and an event of cooling the aerosol may occur. For a diameter of the hole included in the first segment **221**, an appropriate diameter in a range from about 2 mm to about 4.5 mm may be used, but the embodiment is not limited thereto.

[0147] For a length of the first segment **221**, an appropriate length in a range from about 4 mm to about 30 mm may be used, but the embodiment is not limited thereto. For example, the length of the first segment **221** may be about 10 mm, but is not limited thereto.

[0148] A second segment **222** of the filter rod **22** cools the aerosol generated as the heater **182** heats the aerosol generating rod **21**. Accordingly, the user may inhale the aerosol that has been cooled down to an appropriate temperature.

[0149] A length or a diameter of the second segment **222** may be variously determined according to the form of an aerosol generating article **2**. For example, the length of the second segment **222** may be appropriately selected from a range from about 7 mm to about 20 mm. Desirably, the length of the second segment **222** may be about 14 mm, but is not limited thereto.

[0150] The segment **222** may be manufactured by fabricating a polymer fiber. In this case, flavoring liquid may be applied onto the polymer fiber. Alternatively, the second segment **222** may be manufactured by fabricating an additional fiber, onto which flavoring liquid has been applied, and the polymer fiber. Alternatively, the second segment **222** may also be formed by a crimped polymer sheet.

[0151] For example, the polymer sheet may be manufactured with materials selected from a group consisting of polyethylene (PE), polypropylene (PP), polyvinyl carbonate (PVC), polyethylene terephthalate (PET), polylactic acid (PLA), cellulose acetate (CA), and aluminum foil.

[0152] As the second segment **222** is formed from the fabricated polymer fiber or the crimped polymer sheet, the second segment **222** may include a single channel or a plurality of channels extending in a longitudinal direction. Here, the channel(s) may indicate a path (or paths) through which gas (e.g., the air or the aerosol) passes.

[0153] For example, the second segment **222** formed of the crimped polymer sheet may be formed from a material having a thickness about 5  $\mu\text{m}$  to about 300  $\mu\text{m}$ , e.g., about 10  $\mu\text{m}$  to about 250  $\mu\text{m}$ . In addition, the entire surface of the second segment **222** may be between about 300  $\text{mm}^2/\text{mm}$  and about 1,000  $\text{mm}^2/\text{mm}$ . In addition, an aerosol cooling element may be formed from a material having a specific surface area between about 10  $\text{mm}^2/\text{mg}$  and about 100  $\text{mm}^2/\text{mg}$ .

[0154] The second segment **222** may include a thread containing a volatile flavoring ingredient. Here, the volatile flavoring ingredient may include menthol, but is not limited thereto. For example, the thread may be filled with a sufficient amount of menthol to provide 1.5 mg or more menthol to the second segment **222**.

[0155] A third segment **223** of the filter rod **22** according to an embodiment may include a cellulose

acetate filter. A length of the third segment **223** may be appropriately selected from a range from about 4 mm to about 20 mm. For example, the length of the third segment **223** may be about 12 mm, but is not limited thereto.

[0156] The third segment **223** of the filter rod **22** according to an embodiment may include a lyocell filter. A lyocell fiber is an eco-friendly fiber manufactured with cellulose extracted from wood pulp. A lyocell tow indicates a bunch formed by cross-link between lyocell fibers adjacent to each other.

[0157] In some embodiments, the lyocell fiber may have a non-circular cross-section. A non-circular cross-section is defined as a cross-section whose shape is not circular and includes a plurality of protrusions. For example, a cross-section having a shape in which a plurality of protrusions extend from a center may be referred to as a non-circular cross-section.

[0158] In some embodiments, the lyocell fiber may have a Y-shape cross-section having three protrusions branching out from a center, a cross-shape cross-section having four protrusions, a star-shape cross-section having five or more protrusions, or an O-shaped cross-section, but is not limited thereto.

[0159] The capsule **23** to be described below may contain a phenol reduction material. The “phenol” may include a group of chemical compounds consisting of a hydroxyl group ( $\text{—OH}$ ) directly bonded to an aromatic hydrocarbon functional group, and a phenol group includes phenol, catechol, m+P cresol, and o-cresol. The “phenol reduction material” may correspond to a material by which a phenolic material in smoke generated during smoking, e.g., at least one of phenol, catechol, m+P cresol, and o-cresol, may be specifically reduced.

[0160] When the phenol reduction material including polyethylene glycol (PEG), triethyl citrate (TEC), triacetin (TA) and the like are added to cellulose acetate having hydrophobicity, the phenol reduction material may perform a plasticizer function with a cellulose acetate fiber, combine the cellulose acetate fibers having hydrophobicity to each other, and decrease the biodegradability. Here, the term “plasticizer” usually indicates an additive mixed to a polymer to give an elastic modulus and flexibility and decrease melt viscosity of a resin and improve the workability of the resin. However, the expression that the phenol reduction material performs the plasticizer function on the fiber (e.g., cellulose acetate) may also indicate functioning as a binder.

[0161] When the phenol reduction material such as PEG, TEC, or TA is added to the lyocell fiber, the lyocell fiber having hydrophilicity is not plasticized. Accordingly, when the phenol reduction material such as PEG, TEC, or TA is added to the lyocell fiber, the degradability may be not decreased.

[0162] Here, although an embodiment in which the third segment **223** of the filter rod **22** being in directly contact with the capsule **23** to be described later has been described, the embodiment is not limited thereto. For example, the first segment **221** and the second segment **222** may also be manufactured with lyocell filters.

[0163] The filter rod **22** may be manufactured to release flavors. For example, flavoring liquid may be sprayed onto the filter rod **22**. For example, an additional fiber, onto which flavoring liquid has been sprayed, may be inserted into the filter rod **22**.

[0164] Also, the filter rod **22** may include at least one capsule **23**. Here, the capsule **23** may have a double-shell capsule structure, and may include a phenol reduction material. The capsule **23** may be broken by an external force and release the phenol reduction material from inside. The external force may include, for example, a force generated by a hand or a mouth of a smoker, but is not particularly limited thereto. The capsule **23** will be described in further detail with reference to FIG. 7.

[0165] Referring to FIG. 6, the aerosol generating article **3** may further include a front-end plug **33**. The front-end plug **33** may be located on one side of the tobacco rod **31** which is opposite to the filter rod **32**. The front-end plug **33** may prevent the aerosol generating rod **31** from escaping to the outside. The front-end plug **33** may prevent liquefied aerosol from flowing from the aerosol

generating rod **31** into the aerosol generating device **1** during smoking.

[0166] The filter rod **32** may include a first segment **321** and a second segment **322**. The first segment **321** may correspond to the first segment **221** of the filter rod **22** shown in FIG. 5. The second segment **322** may correspond to the third segment **223** of the filter rod **22** shown in FIG. 5. [0167] A diameter and a total length of the aerosol generating article **3** may correspond to a diameter and a total length of the aerosol generating article **2** of FIG. 5. For example, the length of The front-end plug **33** is about 7 mm, the length of the tobacco rod **31** is about 15 mm, the length of the first segment **321** is about 12 mm, and the length of the second segment **322** is about 14 mm, but it is not limited thereto.

[0168] The aerosol generating article **3** may be packaged using at least one wrapper **35**. The wrapper **35** may have at least one hole through which external air may be introduced or internal air may be discharged. For example, the front end plug **33** may be packaged by a first wrapper **351**, the tobacco rod **31** may be packaged by a second wrapper **352**, the first segment **321** may be packaged by a third wrapper **353**, and the second segment **322** may be packaged by a fourth wrapper **354**. Further, the entire aerosol generating article **3** may be repackaged by a fifth wrapper **355**.

[0169] In addition, at least one perforation **360** may be formed in the fifth wrapper **355**. For example, the perforation **36** may be formed in a region surrounding the tobacco rod **31**, but is not limited thereto. For example, the perforation **36** may deliver heat, which is generated by the heat **210** shown in FIG. 3, into the aerosol generating rod **31**.

[0170] In addition, at least one capsule **34** may be included in the second segment **322**. Here, the capsule **34** may have a double-shell capsule structure, and may include a phenol material. The capsule **34** may be broken by an external force and release the phenol reduction material from inside. The external force may include, for example, a force generated by a hand or a mouth of a smoker, but is not particularly limited thereto. The capsule **34** will be described in further detail with reference to FIG. 7.

[0171] A first wrapper **351** may be obtained by combination of a metal foil, e.g., an aluminum foil, to a general type of filter paper. For example, the entire thickness of the first wrapper **351** may be in a range from about 45  $\mu\text{m}$  to about 55  $\mu\text{m}$ . For example, the entire thickness of the first wrapper **351** may be about 50.3  $\mu\text{m}$ . In addition, a thickness of the metal foil of the first wrapper **351** may be in a range from about 6  $\mu\text{m}$  to about 7  $\mu\text{m}$ . For example, the thickness of the metal foil of the first wrapper **351** may be about 6.3  $\mu\text{m}$ . In addition, a basis weight of the first wrapper **351** may be in a range from about 50 g/m.<sup>sup.2</sup> to about 55 g/m.<sup>sup.2</sup>. For example, the basis weight of the first wrapper **351** may be about 53 g/m.<sup>sup.2</sup>.

[0172] The second wrapper **352** and the third wrapper **353** may be manufactured with general filter rolling paper. For example, the second wrapper **352** and the third wrapper **353** may include porous rolling paper or non-porous rolling paper.

[0173] For example, the porosity of the second wrapper **352** may be about 35,000 CU, but is not limited thereto. In addition, a thickness of the second wrapper **352** may be in a range from about 70  $\mu\text{m}$  to about 80  $\mu\text{m}$ . For example, the thickness of the second wrapper **352** may be about 78  $\mu\text{m}$ . In addition, a basis weight of the second wrapper **352** may be in a range from about 20 g/m.<sup>sup.2</sup> to about 25 g/m.<sup>sup.2</sup>. For example, the basis weight of the second wrapper **352** may be about 23.5 g/m.<sup>sup.2</sup>.

[0174] For example, the porosity of the third wrapper **353** may be about 24,000 CU, but is not limited thereto. In addition, a thickness of the third wrapper **353** may be in a range from about 60  $\mu\text{m}$  to about 70  $\mu\text{m}$ . For example, the thickness of the third wrapper **353** may be about 68  $\mu\text{m}$ . In addition, a basis weight of the third wrapper **353** may be in a range from about 20 g/m.<sup>sup.2</sup> to about 25 g/m.<sup>sup.2</sup>. For example, the basis weight of the third wrapper **353** may be about 21 g/m.<sup>sup.2</sup>.

[0175] The fourth wrapper **354** may be manufactured with polylactic acid (PLA) lamination paper. Here, the PLA lamination paper may indicate three-layered paper including a paper layer, a PLA

layer, and a paper layer. For example, a thickness of the fourth wrapper **354** may be in a range from about 100  $\mu\text{m}$  to about 120  $\mu\text{m}$ . For example, the thickness of the fourth wrapper **354** may be about 110  $\mu\text{m}$ . In addition, a basis weight of the fourth wrapper **354** may be in a range from about 80 g/m.<sup>2</sup> to about 100 g/m.<sup>2</sup>. For example, the basis weight of the fourth wrapper **354** may be about 88 g/m.<sup>2</sup>.

[0176] The fifth wrapper **355** may be manufactured with sterile paper (MFW). Here, the sterile paper (MFW) may indicate paper fabricated in a particular manner such that a tensile strength, water resistance, smoothness thereof are improved compared with general paper. For example, a basis weight of the fifth wrapper **355** may be in a range from about 57 g/m.<sup>2</sup> to about 63 g/m.<sup>2</sup>. For example, the basis weight of the fifth wrapper **355** may be about 60 g/m.<sup>2</sup>. In addition, a thickness of the fifth wrapper **355** may be in a range from about 64  $\mu\text{m}$  to about 70  $\mu\text{m}$ . For example, the thickness of the fifth wrapper **355** may be about 67  $\mu\text{m}$ .

[0177] Certain materials may be added to the fifth wrapper **355**. Here, silicon may be an example of the certain materials, but the embodiment is not limited thereto. For example, the silicon may have properties such as heat resistance, i.e., the property of not being easily deformed due to temperatures, oxidation resistance, i.e., the property of not being easily oxidized, resistance against various types of drugs, water-repellency, electrical resistivity, or the like. Although, any material having the aforementioned properties, even the material is not silicon, may be applied (or coated) onto the fifth wrapper **355** without limitation.

[0178] The front-end plug **33** may be manufactured with cellulose acetate. For example, the front-end plug **33** may be manufactured by adding a plasticizer (e.g., triacetin) to a cellulose acetate tow. A mono denier of a filament constructing the cellulose acetate tow may be in a range from about 1.0 to about 10.0. For example, a mono denier of the filament constructing the cellulose acetate tow may be in a range from about 4.0 to about 6.0. For example, a mono denier of a filament of the front-end plug may be about 5.0. In addition, a cross-section of the filament constructing the front-end plug **33** may have a Y shape. A total denier of the front-end plug **33** may be in a range from about 20,000 to about 30,000. For example, the total denier of the front-end plug **33** may be in a range from about 25,000 to about 30,000. For example, the total denier of the front-end plug **33** may be about 28,000.

[0179] In addition, depending on needs, the front-end plug may include at least one channel. A shape of a cross-section of the channel of the front-end plug **33** may be variously formed.

[0180] The aerosol generating rod **31** may correspond to the aerosol generating rod **21** described in detail with reference to FIG. 5. Accordingly, hereinafter, detailed descriptions about the aerosol generating rod **31** will be omitted.

[0181] The first segment **321** may be manufactured with cellulose acetate. For example, the first segment may include a structure having the form of a tube including a hole therein. The first segment **321** may be manufactured by adding a plasticizer (e.g., triacetin) to a cellulose acetate tow. For example, a mono denier and a total denier of the first segment **321** may be identical to the mono denier and the total denier of the front-end plug **33**.

[0182] The second segment **322** may be manufactured with cellulose acetate. A mono denier of a filament constructing the second segment **322** may be in a range from about 1.0 to about 10.0. For example, the mono denier of the filament of the second segment **322** may be in a range from about 8.0 to about 10.0. For example, the mono denier of the filament of the second segment **322** may be about 9.0. In addition, a cross-section of the filament of the second segment may have a Y-shape. A total denier of the second segment **322** may be in a range from about 20,000 to about 30,000. For example, the total denier of the second segment **322** may be about 25,000.

[0183] As described with reference to FIG. 5, the first segment **321** and the second segment **322** of the filter rod **32** may include lyocell filters.

[0184] FIG. 7 is a diagram illustrating a capsule included in the aerosol generating article according to an embodiment. Here, the capsule **23** shown in FIG. 5 and the capsule **34** shown in FIG. 6 have a



substantially same structure. Hereinafter, for convenience of description, FIG. 7 will be described with reference to the capsule 23 in FIG. 5.

[0185] In embodiments, the capsule 23 may include a core region C, a first shell S1, and a second shell S2, and the core region C of the capsule 23 may include the phenol reduction material.

[0186] Referring to FIG. 7, it is illustrated that the capsule 23 is spherical, but the embodiment is not limited thereto, and a cross-section of the capsule 23 may have an oval shape in an area or a circular shape in which a portion is deformed.

[0187] The capsule 23 may be sequentially formed from inside in a double-shell structure including the core region C, the first shell S1, and the second shell S2, and regions may respectively include different materials. For example, the first shell S1 may include an oil-soluble vegetable wax, and the second shell S2 may include a water-soluble polymer.

[0188] The core region C may have a diameter about 2.5 mm to about 6.0 mm, the first shell S1 may have a thickness about 0.1 mm to about 1.0 mm, and the second shell S2 may have a thickness about 0.001 mm to about 1.5 mm. For example, the second shell S2 may have a thickness about 0.3 mm to about 1.5 mm before drying moisture, and may have a thickness about 0.001 mm to about 0.05 mm after the moisture has been dried. However, the embodiment is not limited thereto, and the aforementioned numerical range may be appropriately adjusted at a level easily modified by those skilled in the art.

[0189] The core region C may include a water-soluble phenol reduction material. For example, the phenol reduction material may include PEG. PEG, which includes repeated ether bonds ( $\text{—O—}$ ), may form hydrogen bonds with phenol molecules. A hydroxyl group ( $\text{—OH}$ ) of the phenol molecule and ether oxygen of PEG may be bonded, and therefore, phenol may be stably captured. In addition, PEG, which is a high-molecular weight material, tends to physically adsorb phenol. When PEG adsorbs phenol, the concentration of phenol included in mainstream smoke may be reduced.

[0190] To reduce the phenol component included in the mainstream smoke, the phenol reduction material (e.g., PEG) may be applied onto the filter rod 22 (or the third segment 223). However, during storage of the aerosol generating article 2, the phenol reduction material that has been applied may be depleted, and therefore, the storage stability may be poor. It is required to use the capsule 23 to prevent the phenol reduction material from depletion. However, in a case of a water-soluble material such as PEG, when a capsule only including an existing core region and a water-soluble single shell surrounding the core region, there is possibility that the water-soluble material itself will dissolve the capsule (or the shell). Therefore, it is required that the capsule has a double-shell structure including the core region C, the first shell S1 including the oil-soluble material, and the second shell S2 including the water-soluble material in orders from inside, like the capsule 23 according to an embodiment of the present application.

[0191] According to an embodiment, PEG may have a low molecular weight (that is, in a liquid state). For example, a molecular weight of PEG may be in a range from about 200 to about 600. Desirably, PEG may have a molecular weight of about 400, such that PEG may have viscosity to be uniformly impregnated into the third segment 223 of the filter rod 22 (or the second segment 322 of the filter rod 32) after the capsule 23 is crushed.

#### EXPERIMENT EXAMPLE: MEASUREMENT ON THE QUALITY OF THE CAPSULE ACCORDING TO A MOLECULAR WEIGHT OF A PHENOL REDUCTION MATERIAL

[0192] A plurality of capsules having a same structure as the capsule 23 shown in FIG. 7 have been manufactured with difference molecular weights of PEG included in the core region C, and suitability for capsule manufacturing, suitability for solution absorption, and suitability for phenol reduction of the manufactured capsule have been evaluated. A result of the evaluation was shown in Table 1 below.

[0193] The suitability for capsule manufacturing was evaluated based on the following standards.

[0194] ○: the capsule was easily manufactured [0195] Δ: the capsule was not easily manufactured

[0196] X: the capsule could not be manufactured

[0197] The suitability for solution absorbance was evaluated based on the following standards.

[0198] ○: a phenol reduction material leaked out when the capsule was crushed was uniformly impregnated into a filter [0199] Δ: the phenol reduction material, which leaked out when the capsule was crushed, was not uniformly impregnated into the filter [0200] X: the phenol reduction material, which leaked out when the capsule was crushed, was not impregnated into the filter

[0201] The performance of phenol reduction was evaluated based on the following standards.

[0202] ○: the effect of reducing phenol was excellent [0203] Δ: the effect of reducing phenol was insignificant [0204] X: there was no substantial effect of reducing phenol

TABLE-US-00001 TABLE 1 A molecular weight of the phenol reduction material 100 200 300 400 500 600 700 The suitability for ○ ○ ○ ○ ○ ○ Δ manufacturing the capsule The suitability for ○ ○ ○ ○ ○ ○ Δ solution absorption The performance of X ○ ○ ○ ○ ○ ○ phenol reduction

[0205] As shown in Table 1, when the molecular weight of the phenol reduction material was 700 or higher, the capsule was not appropriately manufactured due to high viscosity. It was found that, when the molecular weight of the phenol reduction material was 700 or higher, due to high viscosity, when the capsule was crushed, the phenol reduction material was not uniformly impregnated into the filter. It was also found that, when the molecular weight of the phenol reduction material was 100 or lower, due to excessively low viscosity, there was no substantial phenol adsorption effect.

[0206] In addition, the core region C may further include a water-soluble flavoring material. The flavoring materials may add flavors to the aerosol generated by the aerosol generating article 2 (see FIG. 5) or the aerosol generating article 3 (see FIG. 6). For example, the water-soluble flavoring material may include at least one selected from a group consisting of esters, alcohols, aldehydes, ketones, phenols, nitrogen-containing compounds, and acids.

[0207] In embodiments, the first shell S1 may include oil-soluble wax having a melting point of about 30° C. to about 60° C., and the second shell S2 may include a water-soluble polymer having a gelation temperature of about 40° C. to about 55° C. More particularly, a melting point of the first shell S1 may be about 30° C. to about 60° C., about 30° C. to about 50° C., about 30° C. to about 40° C., about 40° C. to about 60° C., about 40° C. to about 50° C., or about 50° C. to about 60° C. A melting point of the second shell S2 may be about 45° C. to about 55° C., about 45° C. to about 50° C., or about 50° C. to about 55° C.

[0208] When the melting point of the oil-soluble wax used as the material of the first shell S1 is lower than 30° C., the capsule 23 may be easily broken during storage, and when the melting point of the oil-soluble wax is higher than 60° C., as the capsule 23 has strong mechanical strength, the capsule 23 is not destroyed due to an external force or heat, and therefore, nicotine and the aerosol generating material may be not appropriately released.

[0209] In addition, as the oil-soluble wax used as the material of the first shell S1 has a relatively low melting point of about 60° C. or less, like to be described later, a triple sintered body coextruded from a nozzle may be uniformly cooled within a temperature range that is relatively close to 60° C., accordingly, the first shell S1 and the second shell S2 may be formed together to have appropriate mechanical strength, and the yield of the capsule 23 may increase.

[0210] The first shell S1 may include vegetable wax. The vegetable wax may include, for example, carnauba wax, candelilla wax, castor wax, ouricury wax, and the like. As a desirable embodiment, the first shell S1 may include cocoa butter or shea butter. The first shell S1 may be manufactured by mixing any one or at least two of the aforementioned vegetable wax. When the first shell S1 is manufactured by mixing two or more of the aforementioned vegetable wax, the melting point of the first shell S1 may appropriately change according to physical properties of the vegetable wax being mixed. As the first shell S1 includes the vegetable wax as described above, unpleasant odor generated when the capsule 23 is heated by the aerosol generating device (not shown) may be

reduced.

[0211] In addition, for improvement in the melting point and mechanical properties, the first shell **S1** may further include additives. In desirable embodiments, the additives may include fat acid, and the fat acid may include at least one of palmitic acid, stearic acid, and myristic acid, but is not limited thereto.

[0212] The second shell **S2** may include a water-soluble polymer material. As a component formed at an outermost portion of the capsule **23**, the second shell **S2** may have properties related to the breakage of the capsule **23**. To prevent unintended breakage of the capsule, the second shell **S2** may be formed of a material that is elastic or flexible. The second shell **S2** may include, for example, water-soluble hydrocolloid such as gelatin, agar, carrageenan, alginate, or pectin, gums such as gellan gum, starch such as potato starch or corn starch, and starch derivatives such as dextrin, maltodextrin, and cyclodextrin. In addition, the second shell **S2** may include cellulose derivatives such as hydroxypropyl methyl cellulose (HPMC), hydroxypropyl cellulose (HPC), methyl cellulose (MC), carboxymethyl cellulose (CMC), polyvinyl alcohol, polyol, and the like.

[0213] In embodiments, a total diameter of the capsule **23** may be about 2.6 mm to about 8.5 mm. More particularly, the total diameter of the capsule **23** may be in a range from about 2.6 mm to about 8.5 mm, about 2.6 mm to about 7.5 mm, about 2.6 mm to about 6.5 mm, about 2.6 mm to about 5.5 mm, about 2.6 mm to about 4.5 mm, about 2.6 mm to about 3.5 mm, about 2.6 mm to about 3.0 mm, about 3.0 mm to about 8.5 mm, about 3.0 mm to about 7.5 mm, about 3.0 mm to about 6.5 mm, about 3.0 mm to about 5.5 mm, about 3.0 mm to about 4.5 mm, about 3.0 mm to about 3.5 mm, about 3.5 mm to about 8.5 mm, about 3.5 mm to about 7.5 mm, about 3.5 mm to about 6.5 mm, about 3.5 mm to about 5.5 mm, about 3.5 mm to about 4.5 mm, about 4.5 mm to about 8.5 mm, about 4.5 mm to about 7.5 mm, about 4.5 mm to about 6.5 mm, about 4.5 mm to about 5.5 mm, about 5.5 mm to about 8.5 mm, about 5.5 mm to about 7.5 mm, about 5.5 mm to about 6.5 mm, about 6.5 mm to about 8.5 mm, about 6.5 mm to about 7.5 mm, or about 7.5 mm to about 8.5 mm. However, the diameter of the capsule **23** may be appropriately modified in a range that may be included in a diameter of the aerosol generating article.

[0214] In an embodiment, a crushing strength of the capsule **23** may be about 0.5 kgf to about 3.0 kgf. More particularly, the crushing strength of the capsule **23** may be in a range from about 0.5 kgf to about 3.0 kgf, about 0.5 kgf to about 2.5 kgf, about 0.5 kgf to about 2.0 kgf, about 0.5 kgf to about 1.5 kgf, about 0.5 kgf to about 1.0 kgf, about 1.0 kgf to about 3.0 kgf, about 1.0 kgf to about 2.5 kgf, about 1.0 kgf to about 2.0 kgf, about 1.0 kgf to about 1.5 kgf, about 1.5 kgf to about 3.0 kgf, about 1.5 kgf to about 2.5 kgf, about 1.5 kgf to about 2.0 kgf, about 2.0 kgf to about 3.0 kgf, about 2.0 kgf to about 2.5 kgf, or about 2.5 kgf to about 3.0 kgf. However, the crushing strength of the capsule **23** may be appropriately modified by those skilled in the art within a range where the capsule **23** may be broken by the external force.

[0215] FIG. **8** is a diagram for describing the aerosol generating system **100**. In this case, the capsule **23** in a crushed state may be arranged downstream of the aerosol generating article **2** (or in the filter rod **22**), and the phenol reduction material included in the core region of the crushed capsule **23** may be released and impregnated into the filter rod **22**.

[0216] In embodiments, the aerosol generating device **1** may include the heater **18**, as described with reference to FIG. **1**. The heater **18** may be arranged in the aerosol generating device **1** to surround an accommodation space (not shown) in which the aerosol generating article **2** is accommodated. The heater **18** may, for example, have a cylinder shape surrounding the accommodation space, but is not particularly limited to the aforementioned shape.

[0217] In embodiments, the heater **18** may be arranged to cover at least a portion of the aerosol generating rod **21** of the aerosol generating article **2**. According to a second aspect of the embodiments, a system configured to generate aerosol by heating the aerosol generating article **2** by using the aerosol generating device **1** may be provided.

[0218] Here, the aerosol generating article **2** may include the aerosol generating rod **21**, which

includes the aerosol generating material, and the filter rod **22** arranged downstream of the aerosol generating rod **21** and including the capsule **23**, the capsule **23** may include the core region C, the first shell **S1**, and the second shell **S2**, and the core region may include the phenol reduction material. The aerosol generating device **1** may generate aerosol by heating at least a portion of the aerosol generating rod **21**.

[0219] The filter rod **22** according to an embodiment may include a lyocell filter. A lyocell fiber is an eco-friendly fiber manufactured with cellulose extracted from wood pulp. A lyocell tow indicates a bunch formed by cross-link between lyocell fibers adjacent to each other.

[0220] In some embodiments, the lyocell fiber may have a non-circular cross-section. A non-circular cross-section is defined as a cross-section whose shape is not circular and includes a plurality of protrusions. For example, a cross-section having a shape in which a plurality of protrusions extend from a center may be referred to as a non-circular cross-section.

[0221] In some embodiments, the lyocell fiber may have a Y-shape cross-section having three protrusions branching out from a center, a cross-shape cross-section having four protrusions, a star-shape cross-section having five or more protrusions, or an O-shaped cross-section, but is not limited thereto.

[0222] The capsule **23** may include a phenol reduction material. For example, the phenol reduction material may include polyethylene glycol.

[0223] According to an embodiment, PEG may have a low molecular weight (that is, in a liquid state). For example, a molecular weight of PEG may be in a range from about 200 to about 600. Desirably, PEG may have a molecular weight of about 400, such that PEG may have viscosity to be uniformly impregnated into the third segment **223** of the filter rod **22** (or the second segment **322** of the filter rod **32**) after the capsule **23** is broken.

[0224] In addition, the core region C may further include a water-soluble flavoring materials. The flavoring materials may add flavors to the aerosol generated by the aerosol generating article **2** (see FIG. 5) or the aerosol generating article **3** (see FIG. 6). For example, the water-soluble flavoring material may include at least one selected from a group consisting of esters, alcohols, aldehydes, ketones, phenols, nitrogen-containing compounds, and acids.

[0225] As described above, according to the aerosol generating system **100** in an embodiment, by providing a water-soluble phenol reducing material (e.g., polyethylene glycol) by using the capsule **23**, the storage stability of the aerosol generating article **2** may be secured, and by using the filter rod **22** including the lyocell tow, a decrease in the biodegradability of the aerosol generating system may be prevented.

[0226] Some or other embodiments described above are not exclusive to or distinguished from each other. In some or other embodiments described above, components or functions thereof may be used together or combined.

[0227] For example, component A described with reference to a particular embodiment and/or drawing and component B described with reference to another embodiment and/or drawing may be combined to each other. In other words, even when the combination between components is not directly described, except when the combination is impossible, the components may be made.

[0228] Detailed descriptions given above will be considered only as illustrative, not as limitations. The scope of the disclosure will be determined based on reasonable interpretation of the following claims, and any modifications within a range of equivalents of the disclosure will be included in the scope of the disclosure.

[0229] The disclosure is to provide the aerosol generating article by which ingredients such as phenols in ingredients of cigarette smoke during smoking may be more efficiently removed by containing a phenol reduction material in a capsule having a double-shell structure and securing the storage stability.

[0230] Advantageous effects of the embodiments are not limited to the aforementioned effects, and

unmentioned effects may be clearly understood to those skilled in the art from the present specification and the accompanying drawings.

## Claims

1. An aerosol generating article comprising: an aerosol generating rod comprising an aerosol generating material; and a filter rod arranged downstream of the aerosol generating rod and comprising a capsule, wherein the capsule comprises a core region, a first shell, and a second shell, and the core region comprises a phenol reduction material.
  2. The aerosol generating article of claim 1, wherein the phenol reduction material comprises polyethylene glycol that is a water-soluble material.
  3. The aerosol generating article of claim 2, wherein the polyethylene glycol is in a liquid state, and a molecular weight of the polyethylene glycol is in a range from about 200 to about 600.
  4. The aerosol generating article of claim 1, wherein the core region further comprises a water-soluble flavoring material.
  5. The aerosol generating article of claim 1, wherein the first shell comprises an oil-soluble material, and the second shell surrounds the first shell and comprises a water-soluble material.
  6. The aerosol generating article of claim 5, wherein the first shell comprises one or more of carnauba wax, candelilla wax, castor wax, ouricury wax, cocoa butter, and shea butter.
  7. The aerosol generating article of claim 5, wherein the second shell comprises one or more water-soluble polymers selected from a group consisting of gelatin, agar, carrageenan, gellan gum, pectin, starch, and alginate.
  8. The aerosol generating article of claim 1, wherein a diameter of the capsule is about 2.6 mm to about 8.5 mm, and a crushing strength of the capsule is about 0.5 kgf to about 3.0 kgf.
  9. The aerosol generating article of claim 1, wherein the filter rod comprises a lyocell tow comprising a plurality of lyocell fibers.
  10. The aerosol generating article of claim 9, wherein each of the plurality of lyocell fibers has a non-circular cross-section.
  11. The aerosol generating article of claim 9, wherein the phenol reduction material does not comprise a material for combination among the plurality of lyocell fibers.
  12. The aerosol generating article of claim 1, wherein the aerosol generating article comprises a wrapper covering the aerosol generating rod and the filter rod.
  13. An aerosol generating system comprising: an aerosol generating article comprising an aerosol generating rod and a filter rod, wherein the aerosol generating rod comprises an aerosol generating material and the filter rod is arranged adjacent to the aerosol generating rod and comprises a capsule; and an aerosol generating device comprising an insertion space and a heater, wherein the insertion space is used for accommodating the aerosol generating article and the heater is arranged to correspond to the aerosol generating rod around the insertion space, wherein the capsule comprises a core region, a first shell, and a second shell, and the core region comprises a phenol reduction material.
  14. The aerosol generating system of claim 13, wherein the phenol reduction material comprises polyethylene glycol that is a water-soluble material.
  15. The aerosol generating system of claim 14, wherein the polyethylene glycol is in a liquid state, and a molecular weight of the polyethylene glycol is in a range from about 200 to about 600.
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